

Google, Intel and Mirantis sketch out future of OpenStack

OpenStack distributor Mirantis has partnered with Google and Intel to take the cloud platform to the next level. The new partnership is aimed at packaging OpenStack into Docker containers by leveraging container cluster manager Kubernetes.

Container support is not new for OpenStack. However, the latest collaboration is designed to give users precise control over the placement of services used for the OpenStack control plane. The new development will also provide a 'self-healing' experience on the OpenStack control plane and open new avenues for Kubernetes-powered cloud solutions such as Google Cloud.

"With the emergence of Docker as the standard container image format and Kubernetes as the standard for container orchestration, we are finally seeing continuity in how people approach operations of distributed applications. Combining Kubernetes and Fuel will open OpenStack up to a new delivery model that allows faster consumption of updates, helping customers get to outcomes faster," said Boris Renski, Mirantis CEO, in a statement.

Google believes its tie-up with Mirantis and Intel will bridge the gap between legacy infrastructure software and the next-generation of application development.

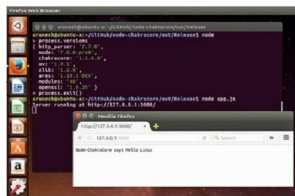
"Many enterprises will benefit from using containers and sophisticated cluster management as the foundation for resilient, highly scalable infrastructure," said Craig McLuckie, Google's senior product manager.

The trio is set to utilise the OpenStack Fuel deployment tool to empower Kubernetes. The tool will reduce the burden of running Docker containers on top-end Linux hardware. Additionally, it will bring Kubernetes into the limelight for various OpenStack developments.

The results of the new announcement are expected to be ready in 2017.

fundamental goals of this project from the beginning has been to ensure that the existing ecosystem continues to work, in an open and cross-platform way, exactly like Node.js," sources in the company added.

Microsoft is updating its ChakraCore roadmap to enhance its cross-platform support. Moreover, the software giant is in development to bring a 'fully-capable'



ChakraCore JIT compiler and Windows-exclusive concurrent and partial GC to other platforms. These new efforts are likely to improve the performance of Node.js and other applications hosting the ChakraCore engine.

Developers have started using ChakraCore for their projects on GitHub. These developments help Microsoft fix and advance the engine for multiple computing platforms.

Google tweaks Linux kernel in Android to enhance security

Google has enabled new protection features on the Linux kernel within Android to increase the security of the open source platform. The features are classified into two different categories—memory protection and attack surface reduction.

"Android relies on the Linux kernel for enforcement of its security model. To better protect the kernel, we've enabled a number of mechanisms within Android," Android security team member Jeff Vander Stoep wrote in a blog post. Among the new changes, there is memory protection for user space processes through address space separation. This tweak enables the Linux kernel to maintain its integrity despite some vulnerabilities within the unrelated portions of the system's memory. Google has also provided a feature to segment kernel memory into logical sections and sets.

To prevent direct access to user space memory, the kernel now has restricted access. "This can make a number of attacks more difficult because attackers have significantly less control over kernel memory that is executable," Stoep said.

Google has also reduced the attack surface to expose fewer entry points to the kernel. For this, the operating system now has the kernel's perf system. Stoep revealed that Android Nougat will by default block access to perf. However, there will be an option to use the perf system via developer settings.

The kernel additionally restricts app access to ioctl commands. These commands previously helped attackers gain backdoor access to Android. Although developers do not use most of the ioctl commands, some third-party apps access them. Thus, Android Nougat will carry a precise whitelist of socket ioctl commands. The new operating system will also come with seccomp to enable an additional sandboxing mechanism.

"Due to these efforts and others, we expect the security of the kernel to

